

AI-POWERED INTELLIGENT EDUCATION SUPPORT SYSTEM



PROBLEMS IN BRIEF

Current Challenges in Education Monitoring

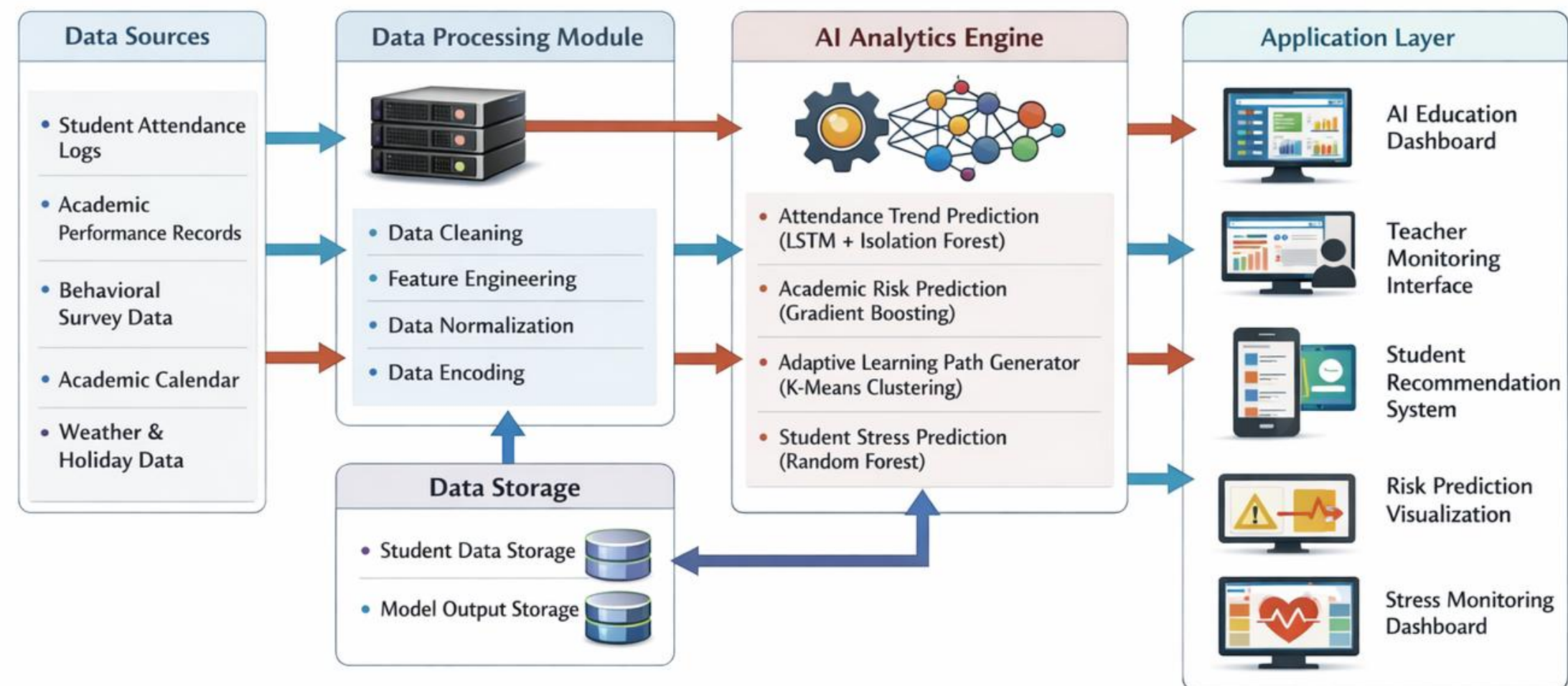
- Students face difficulty identifying academically struggling early.
- Student attendance patterns are often ignored until academic performance significantly drops.
- Students experience increasing academic pressure during O/L examinations.
- Lack of personalized learning recommendations for students with weak subject performance.
- Existing education monitoring systems rely heavily on manual observation rather than intelligent analytics.

SOLUTION OVERVIEW

- **Attendance Trend Analysis** – Analyzes historical attendance records to identify declining attendance patterns using time-series prediction models.
- **Academic Risk Prediction** – Uses machine learning models to classify students into Low, Medium, and High risk levels to identify struggling students early.
- **Adaptive Learning Path Generator** – Analyzes student academic performance and recommends personalized learning priorities for subjects that need improvement.
- **Student Stress Prediction** – Evaluates behavioral factors such as study hours, attendance trends, and academic performance to estimate student stress levels and provide support recommendations.

SYSTEM OVERVIEW DIAGRAM

Figure 1: EduGuide – AI Powered Intelligent Education Support System



IT22907820 | R.P.A.P PRABUDDHINI



ATTENDANCE TRENDS ANALYZER

SPECIALIZATION : INFORMATION TECHNOLOGY



ATTENDANCE TRENDS ANALYZER

How the Solution Addresses the Sub-Problem/ Prototype

- Uses ARIMA/LSTM models to forecast attendance trends.
- Detects anomalies (e.g., sudden class-wide absenteeism) with contextual explanation
- Correlates drops with Sri Lankan realities: weather, holidays
- Delivers prescriptive insights

User Requirements Addressed

- Students - Transparency: See their own attendance trend. Awareness: Understand how weather/exams affect class attendance. Empowerment: Self-monitor before falling at-risk
- Teachers - Real-time alerts, student risk reports, intervention recommendations
- Parents - (Future) Notifications for chronic absenteeism

ATTENDANCE TRENDS ANALYZER

Design Excellence / Contribution

- Predictive Attendance Forecasting - LSTM models to predict future attendance trends and identify early risk patterns.
- Automated Anomaly Detection - Isolation Forest to flag students with irregular or abnormal attendance behavior.
- Context-Aware Analysis - Correlates attendance with exams, holidays, weather
- Actionable Visual Analytics - Provides dashboards, trend charts, and heatmaps for data-driven decision-making

User Feedback on Prototype

- Trend-based model instead of traditional static attendance checking
- Simple, interactive dashboard for quick insight interpretation.
- Modular design can be integrated with other academic analysis systems.
- Focuses on both individual student patterns and class-wide attendance behavior.

PROJECT EVIDENCE

```
print(f"[SECTION 1] Preparing Data for Grade 11 Prediction...")

# 1. Filter df_attendance for students in Grade 10 during 2023
df_grade_10_2023 = df_attendance[(df_attendance['grade'] == 10) & (df_attendance['date'].dt.year == 2023)]
print(f"- Identified {len(df_grade_10_2023)} attendance records for students in Grade 10 in 2023.")

# 2. Extract unique student IDs from df_grade_10_2023
grade_10_students_2023 = df_grade_10_2023['student_id'].unique()
print(f"- Found {len(grade_10_students_2023)} unique students who were in Grade 10 in 2023.")

# 3. Generate a date range for the entire year 2024
start_date_2024 = datetime(2024, 1, 1)
end_date_2024 = datetime(2024, 12, 31)
date_range_2024 = pd.date_range(start_date_2024, end_date_2024, freq='D')
print(f"- Generated a date range for 2024 from {start_date_2024} to {end_date_2024}.")

# 4. Filter this date range to include only weekdays (Monday to Friday)
school_days_2024 = [d for d in date_range_2024 if d.weekday() < 5]
print(f"- Identified {len(school_days_2024)} school days in 2024.")

# 5. Initialize an empty list for Grade 11 2024 data
grade_11_2024_data = []

# 6. Iterate through each unique student ID and each school day in 2024
for student_id in grade_10_students_2023:
    for date in school_days_2024:
        grade_11_2024_data.append({
            'student_id': student_id,
            'date': date,
            'attended': None, # To be predicted
            'grade': 11,
            'day_of_week': date.weekday(),
            'month': date.month,
            'is_exam_month': 0, # Unknown for prediction year
            'is_season': 0 # Unknown for prediction year
        })

# 7. Convert the list into a new Pandas DataFrame
df_grade_11_2024_prediction = pd.DataFrame(grade_11_2024_data)
print(f"- Created prediction DataFrame with {len(df_grade_11_2024_prediction)} records for Grade 11 in 2024.")
```

```
print(f"[SECTION 2] Exploratory Data Analysis...")

# Summary statistics
print(f"--- Attendance Summary by Grade ---")
grade_summary = df_attendance.groupby('grade')[['attended']].agg(['mean', 'sum', 'count'])
print(grade_summary)

# Monthly trends
monthly_attendance = df_attendance.groupby('month')['attended'].mean()
print(monthly_attendance)

# Visualization
fig, ax = plt.subplots(1, 2, figsize=(15, 10))

# Plot 1: Overall attendance trend over time
df_daily = df_attendance.groupby('date')['attended'].mean()
ax[0].plot(df_daily.index, df_daily.values, linewidth=1.5, color='teal')
ax[0].set_title('Overall Daily Attendance Rate (2023-2024)')
ax[0].set_xlabel('Date')
ax[0].set_ylabel('Attendance Rate')

# Plot 2: Monthly attendance pattern
ax[1].bar(monthly_attendance.index, monthly_attendance.values, color='coral')
ax[1].set_title('Average Attendance by Month')
ax[1].set_xlabel('Month')
ax[1].set_ylabel('Attendance Rate')

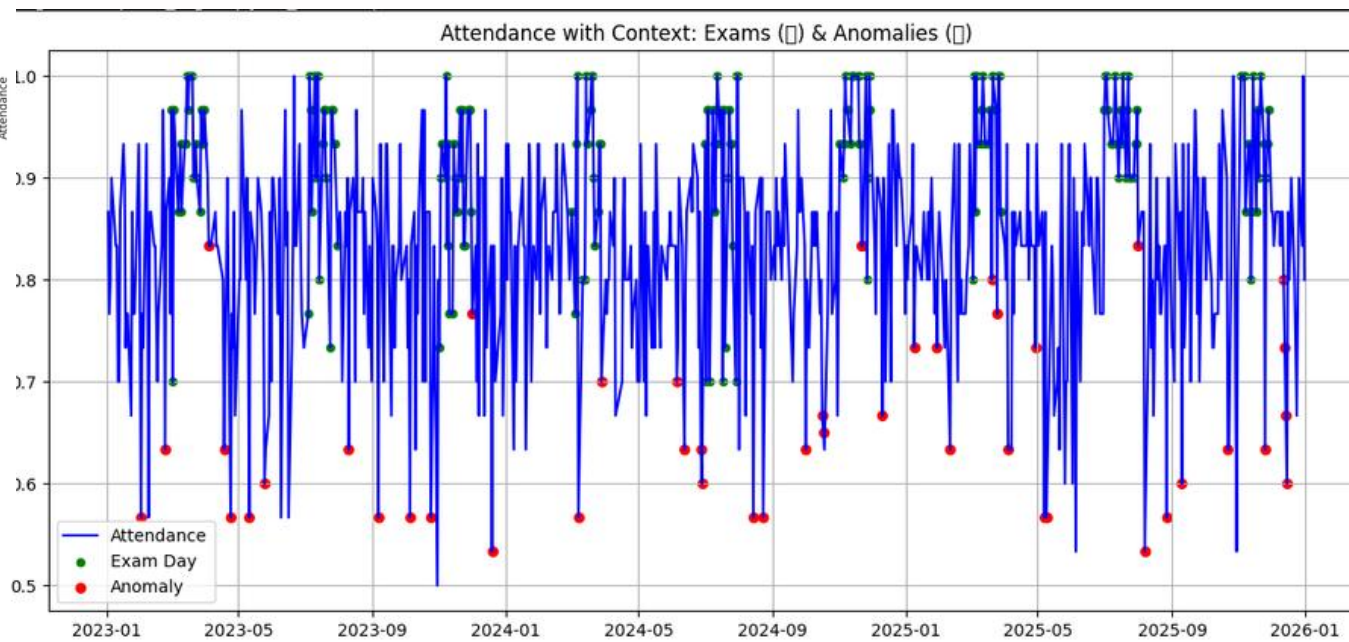
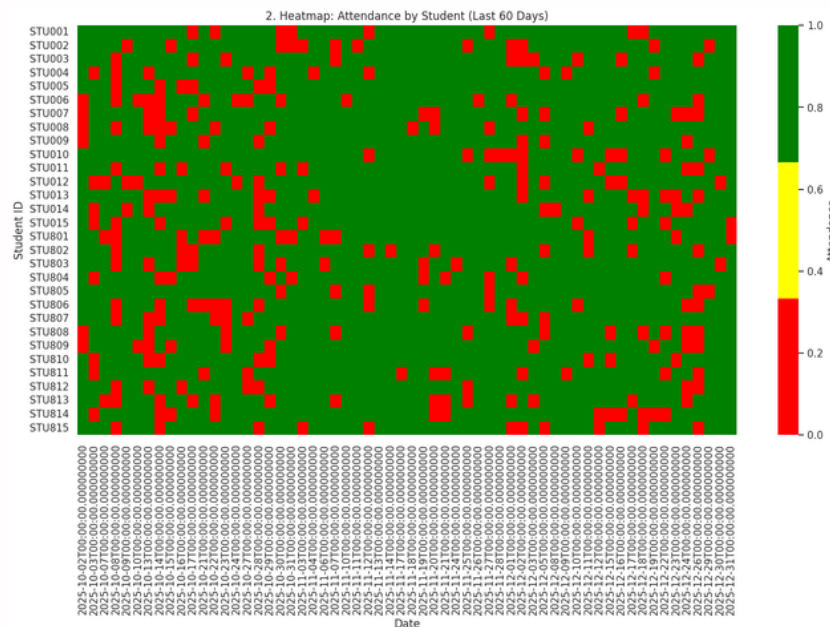
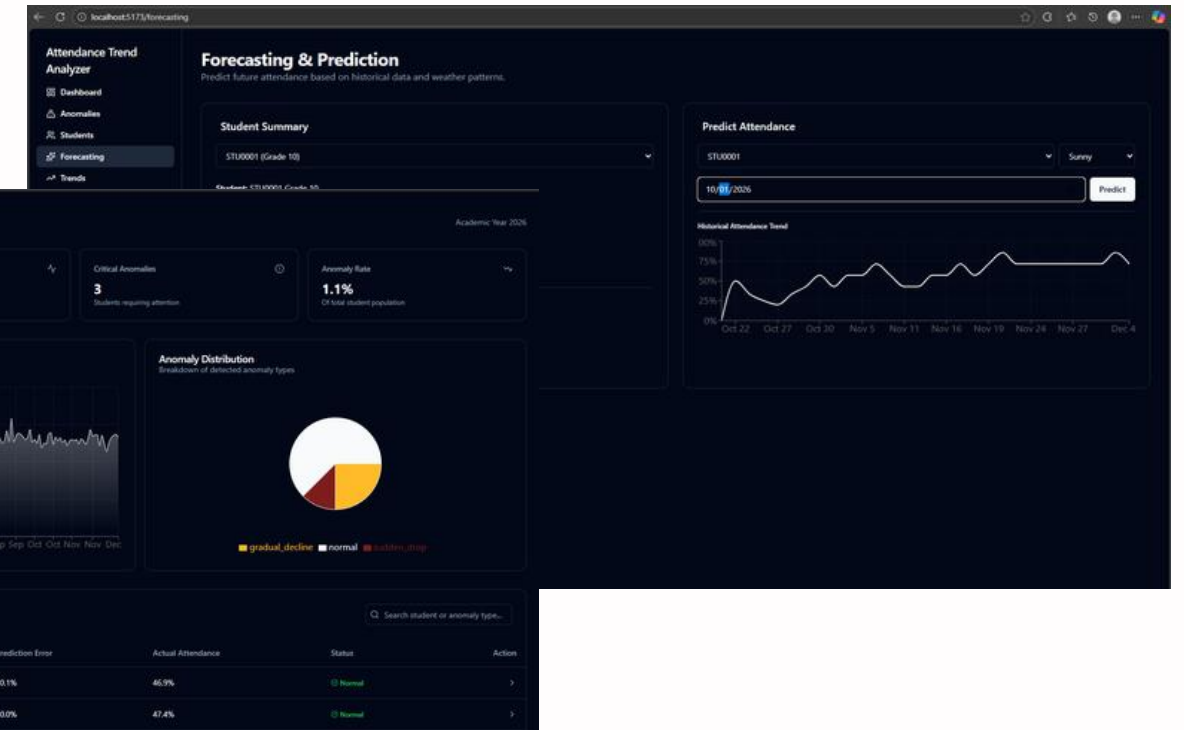
# Reasoning: Now, calculate the cumulative average attendance for each student in df_attendance to serve as a historical attendance feature. Then, create a lagged attendance feature by shifting the 'attended' column by one day, grouped by 'student_id', which will represent the attendance on the previous school day.
```

```
print(f"[SECTION 4] Feature Engineering for Modeling...")

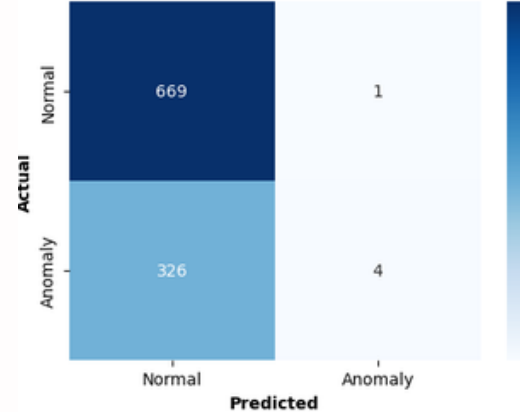
# 1. Extract 'week of year' and 'day of year' for df_attendance
df_attendance['week_of_year'] = df_attendance['date'].dt.isocalendar().week.astype(int)
df_attendance['day_of_year'] = df_attendance['date'].dt.dayofyear
print(f"- Extracted week and day of year for df_attendance.")

# 1. Extract 'week of year' and 'day of year' for df_grade_11_2024_prediction
df_grade_11_2024_prediction['week_of_year'] = df_grade_11_2024_prediction['date'].dt.isocalendar().week.astype(int)
df_grade_11_2024_prediction['day_of_year'] = df_grade_11_2024_prediction['date'].dt.dayofyear
print(f"- Extracted week and day of year for df_grade_11_2024_prediction.")

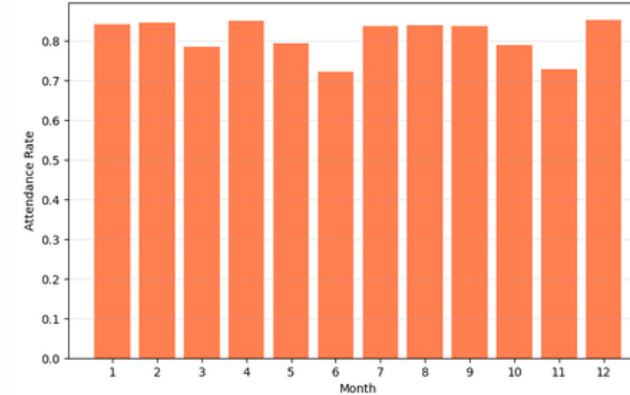
print(f"df_attendance head with new features:")
print(df_attendance[['date', 'week_of_year', 'day_of_year']].head())
print(f"df_grade_11_2024_prediction head with new features:")
print(df_grade_11_2024_prediction[['date', 'week_of_year', 'day_of_year']].head())
```



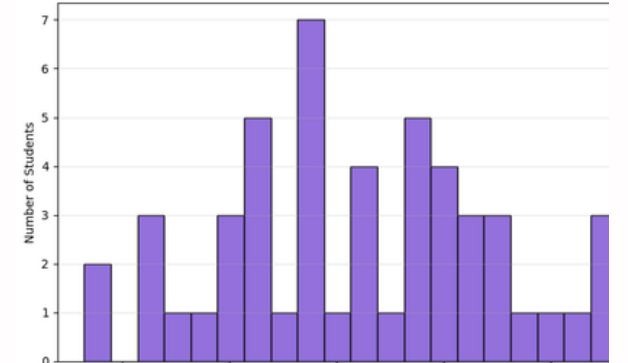
Anomaly Detection Confusion Matrix



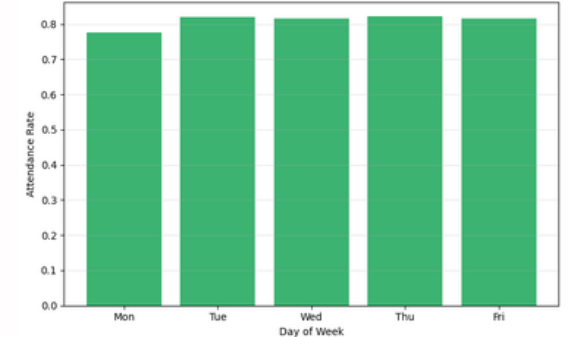
Average Attendance by Month



Distribution of Student Attendance Rates



Attendance by Day of Week



IT22590626 | B.P.D.M.K.ARIYATHILAKA



ACADEMIC RISK PREDICTION SYSTEM

SPECIALIZATION : INFORMATION TECHNOLOGY

ACADEMIC RISK PREDICTION SYSTEM

How the Solution Addresses the Sub-Problem/ Prototype

- Predicts students' academic risk levels using machine learning.
- Identifies early warning signs through attendance, marks, behavioral trends, and performance patterns.
- Supports teachers Students with data-driven decision making.
- Reduces dropout risk and improves student performance through early intervention.

User Requirements Addressed

- Ability to classify students as High / Medium / Low Risk
- Simple, understandable dashboard and visual outputs
- Quick and accurate automated reporting
- Actionable insights for teachers and students

ACADEMIC RISK PREDICTION SYSTEM

Design Excellence / Contribution

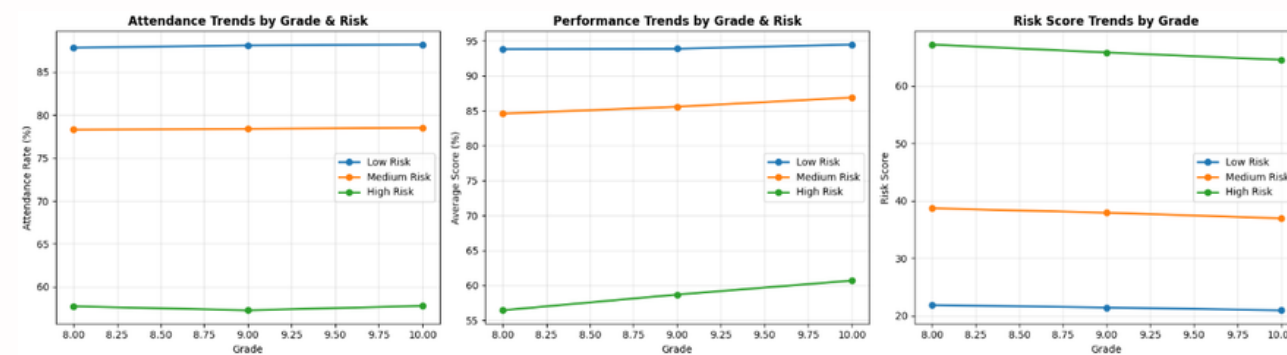
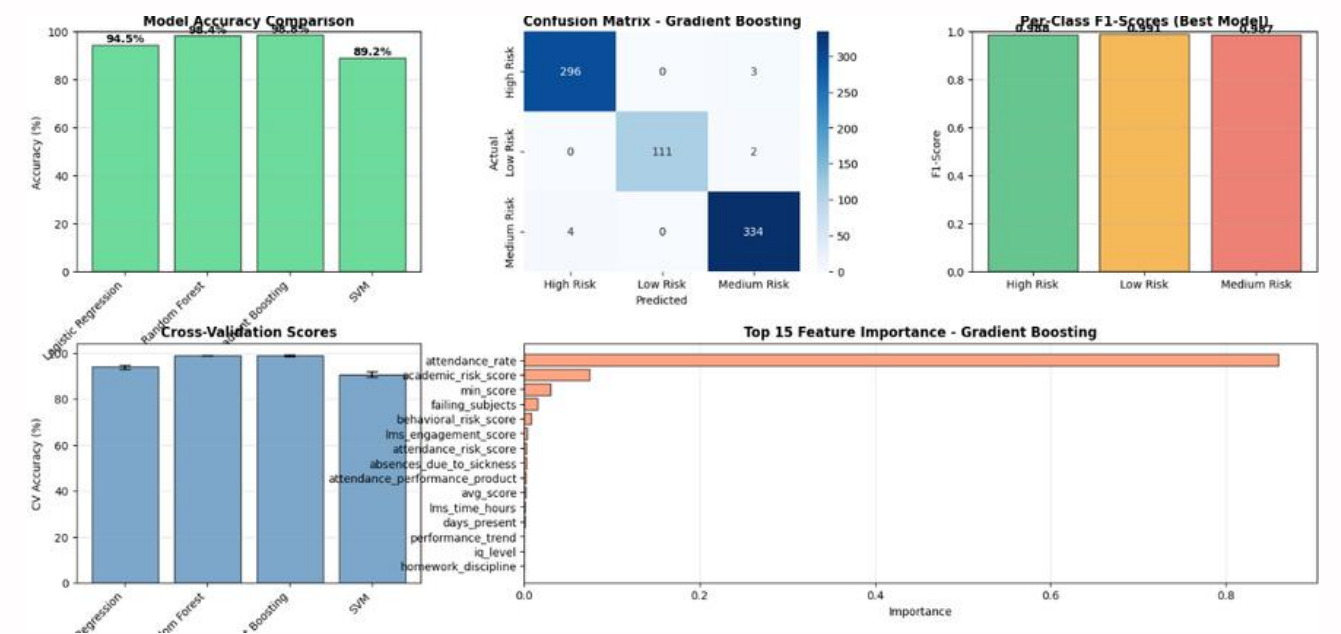
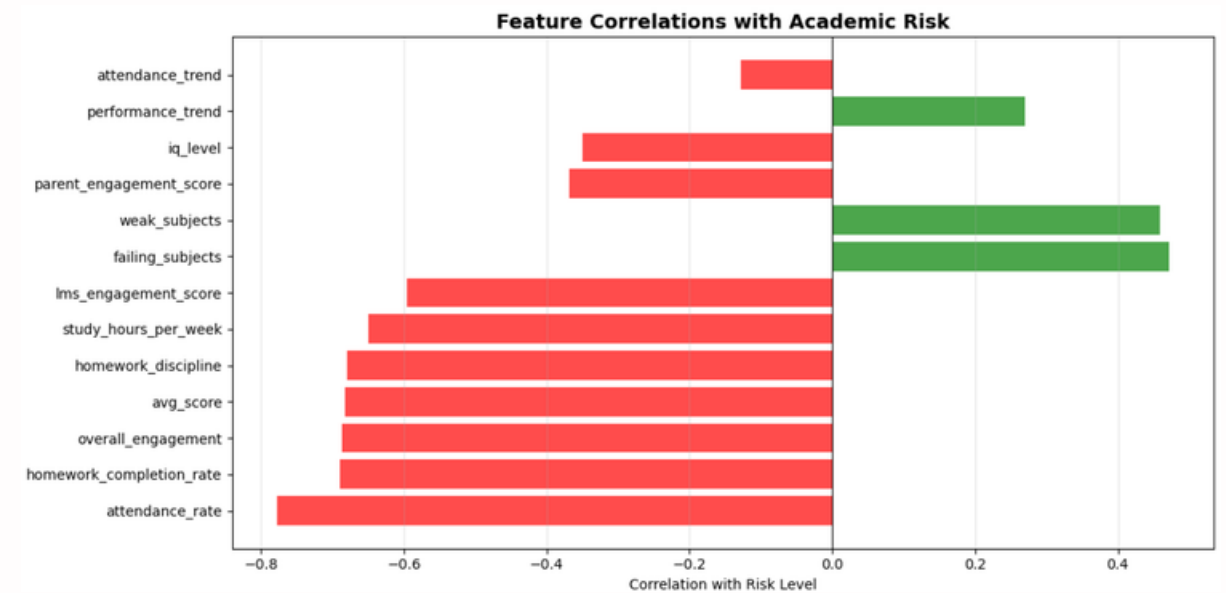
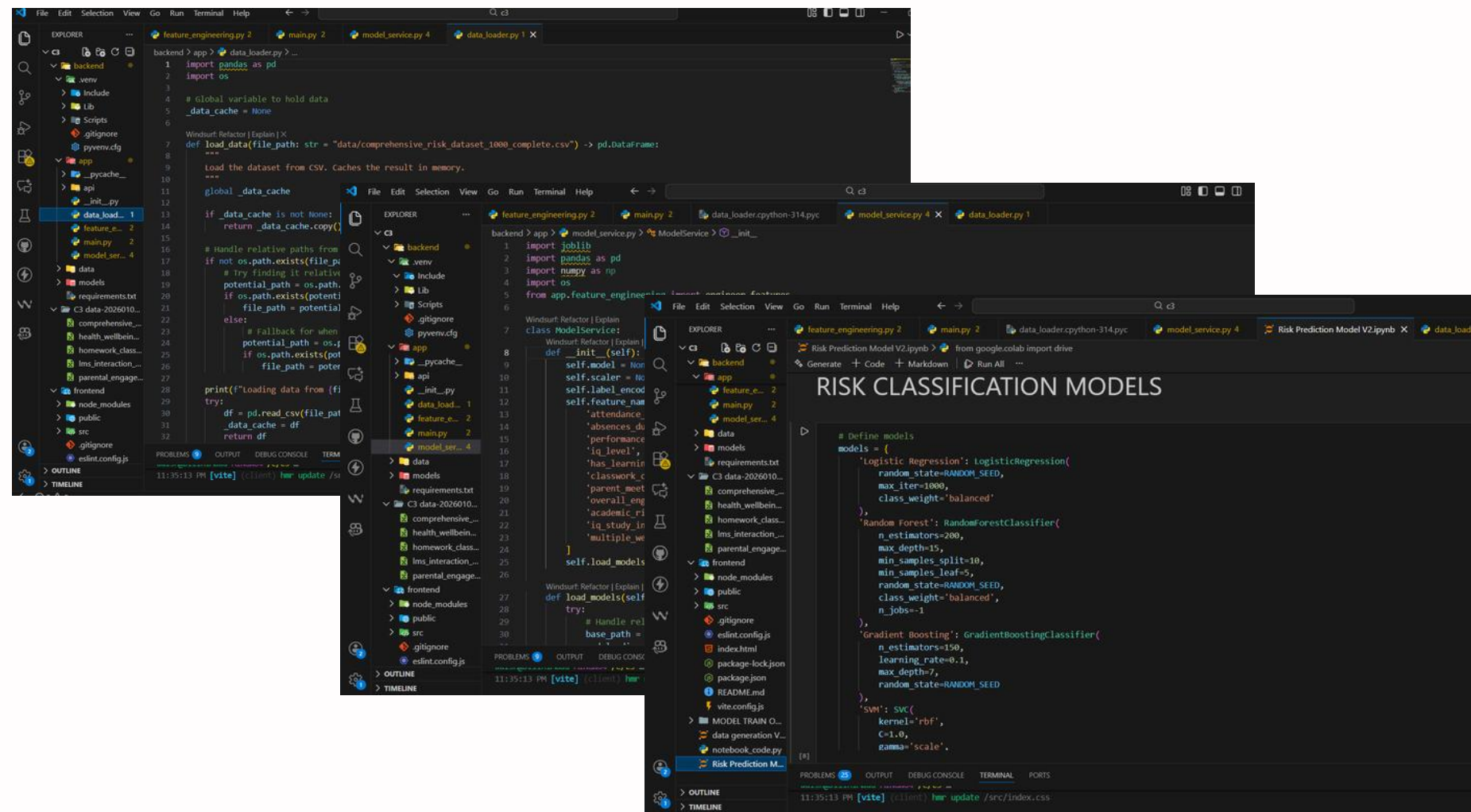
- Uses advanced ML techniques with best-model selection for high accuracy
- Well-structured modular system scalable and adaptable
- User-friendly UI with meaningful visual analytics
- Provides interpretable outputs instead of predictions

User Feedback on Prototype

- Suggested future enhancements: real-time data updates and more behavioral metrics.
- Clear visual indicators made it easy to understand risk levels.



PROJECT EVIDENCE



IT22565730 | K.M.S.DULSHAN



ADAPTIVE LEARNING PATH GENERATOR

SPECIALIZATION : INFORMATION TECHNOLOGY



ADAPTIVE LEARNING PATH GENERATOR

How the Solution Addresses the Sub-Problem/ Prototype

- The solution uses AI-Based Adaptive Learning to analyze student performance data.
- A personalized learning path is generated instead of a one-size-fits-all approach.
- The system automatically identifies weak subjects and learning gaps of each student.
- Suitable learning resources, guidance, and recommendations are provided according to each learner's level.
- Students are grouped into clusters (High, Upper-Average, Average, Low) based on marks and learning patterns.

User Requirements Addressed

- Students receive personalized learning guidance
- System identifies weak subjects automatically
- Provides relevant learning resources
- Simple & user-friendly interface
- Improves learning outcomes through data-driven recommendations

ADAPTIVE LEARNING PATH GENERATOR

Design Excellence / Contribution

- Uses K-Means Clustering + Rule-based logic for accurate grouping
- Adaptive learning approach rather than one-size-fits-all
- Clear visualization of student categories and learning progress
- Scalable and flexible to integrate with school LMS platforms

User Feedback on Prototype

- Users highlighted clear output & meaningful suggestions
- Students found the recommendations useful and easy to follow
- Teachers appreciated automated performance analysis
- Feedback indicated increased motivation and guidance clarity



PROJECT EVIDENCE

VS Code interface showing the project structure and code for the Adaptive Learning Path Generator component. The Explorer panel on the left shows the file structure, including folders for 'src', 'components', and 'data'. The main editor displays the 'student.py' file, which contains code for generating learning paths. The code includes imports for 'google.colab', 'pandas', 'numpy', 'matplotlib.pyplot', 'sklearn', and 'warnings'. The code also includes a function to generate learning paths based on student data.

```
from google.colab import drive
drive.mount('/content/drive')

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score, davies_bouldin_score
import warnings
warnings.filterwarnings('ignore')
```

CONFIGURATION

VS Code interface showing the project structure and code for the Dashboard component. The Explorer panel on the left shows the file structure, including folders for 'src', 'components', and 'data'. The main editor displays the 'Dashboard.jsx' file, which contains code for rendering the dashboard. The code includes imports for 'React', 'useEffect', 'useState', 'react-chartjs-2', 'react-router-dom', and 'react-toastify'. The code also includes a function to fetch student data from the API.

```
import React, { useEffect, useState } from 'react';
import { BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip, Legend, ResponsiveContainer, PieChart, Pie, Cell } from 'react-chartjs-2';
import { Users, BookOpen, Activity, Award } from 'lucide-react';
import { client } from '../api/client';
import { Link } from 'react-router-dom';

const COLORS = ['#3b82f6', '#8b5cf6', '#a0629d', '#f1c40f'];

const Dashboard = () => {
  const [students, setStudents] = useState([]);
  const [loading, setLoading] = useState(true);

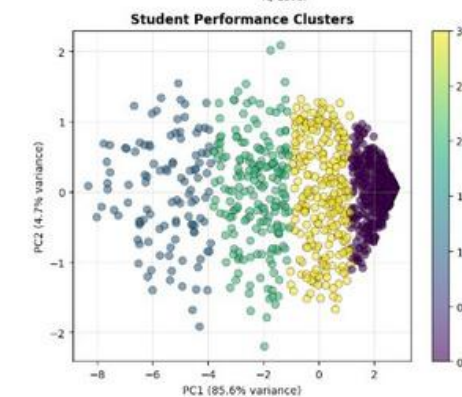
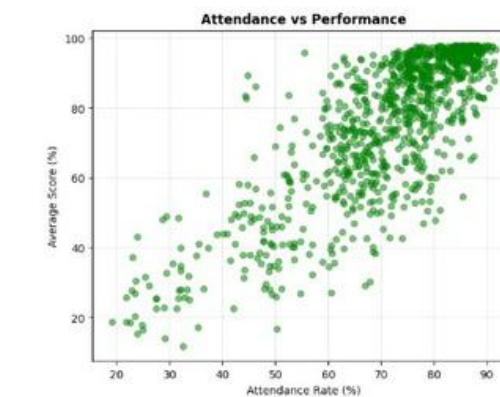
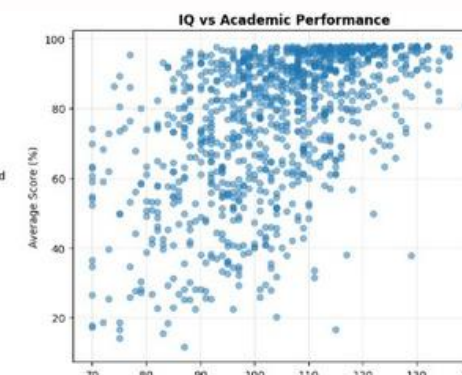
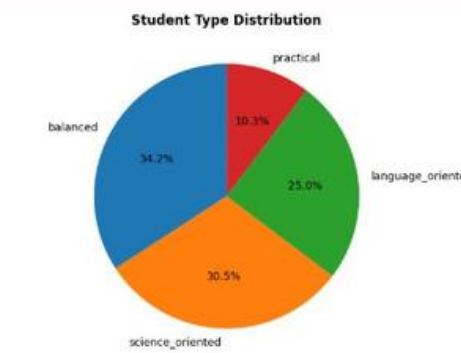
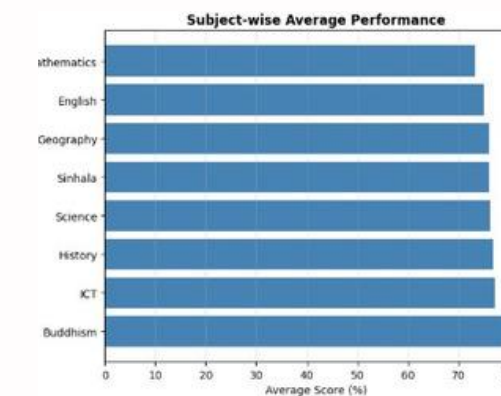
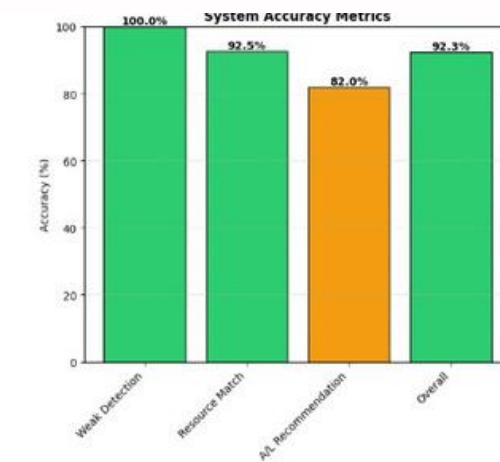
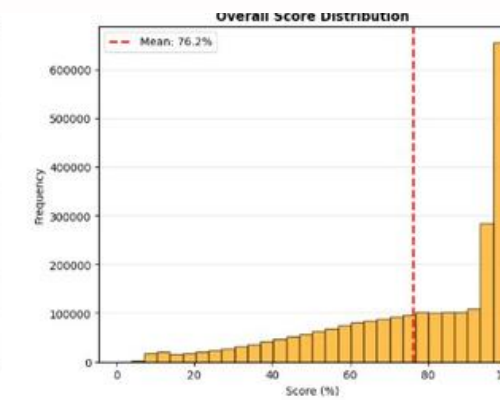
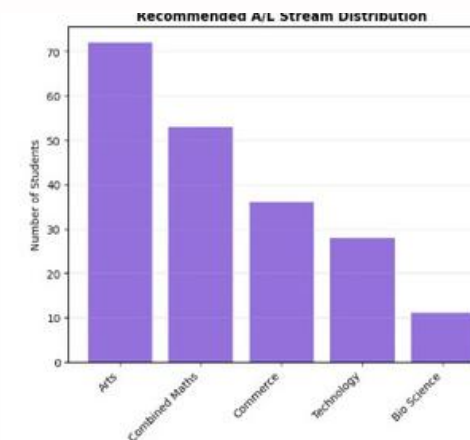
  useEffect(() => {
    const fetchStudents = async () => {
      try {
        const response = await client.get('/students');
        setStudents(response.data);
      } catch (error) {
        console.error("Error fetching students:", error);
      } finally {
        setLoading(false);
      }
    };

    fetchStudents();
  }, []);

  if (loading) return <div className="text-center mt-8">loading dashboard...</div>;

  // Process data for charts
  const clusterCounts = students.reduce((acc, curr) => {
    acc[curr.cluster] = (acc[curr.cluster] || 0) + 1;
    return acc;
  }, {});

  const pieData = Object.keys(clusterCounts).map(key => ({
    name: `cluster ${key}`,
```



IT22571984 | H P R D T Wickramarathna



STRESS LEVEL PREDICTION

SPECIALIZATION : INFORMATION TECHNOLOGY



STRESS LEVEL PREDICTION

How the Solution Addresses the Sub-Problem/ Prototype

- Identifies student stress early using machine learning rather than manual observation.
- Predicts stress risk level (Low / Moderate / High) based on academic, lifestyle and external factors.
- Detects key stress causes such as sleep deprivation, tuition overload, poor attendance and financial pressure.
- Provides AI-powered personalized recommendations instead of generic advice.

User Requirements Addressed

- Students can easily check their stress level through a simple web interface.
- System explains why stress is predicted, not only the result.
- Provides practical and personalized wellbeing guidance.
- Protects all personal student data through privacy-first design.

STRESS LEVEL PREDICTION

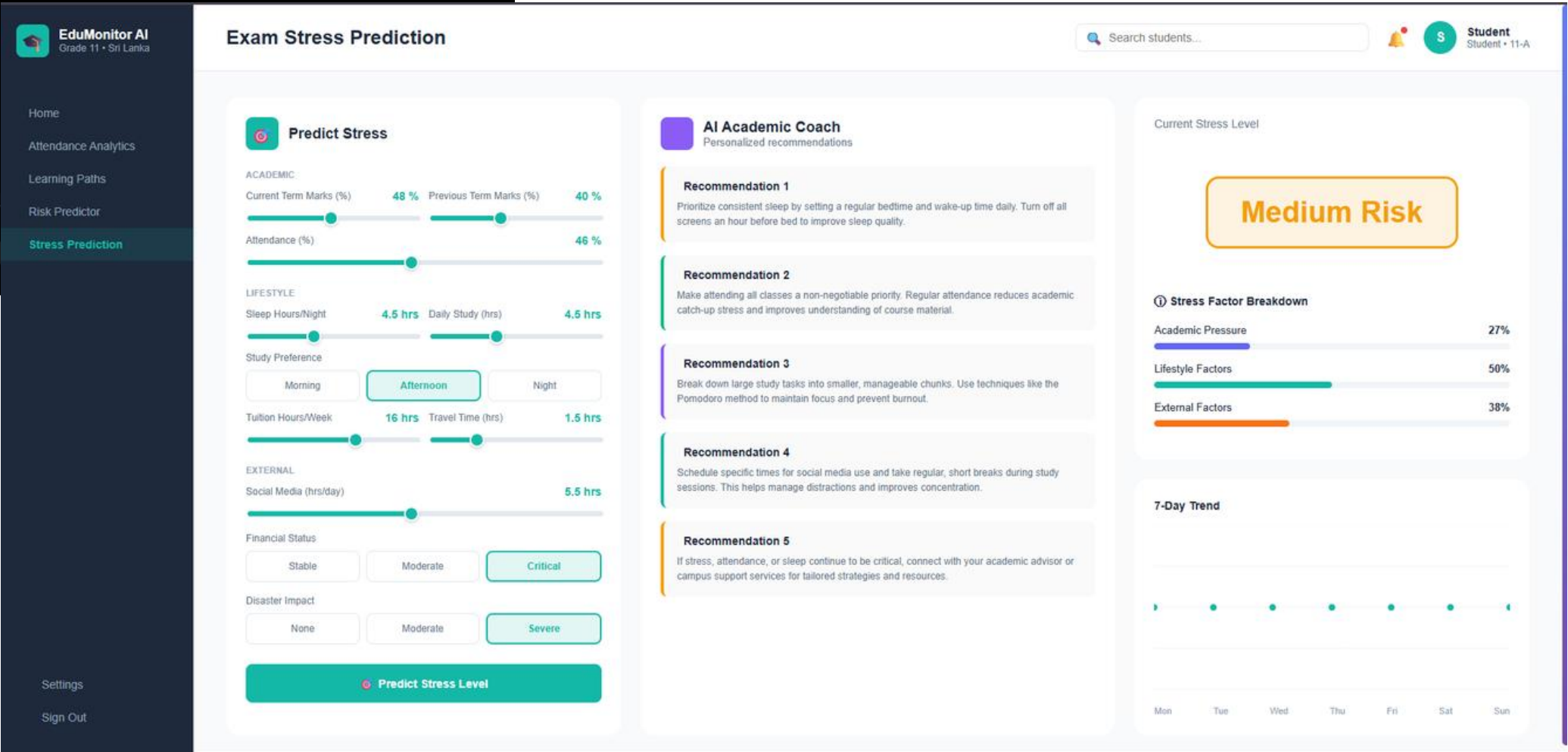
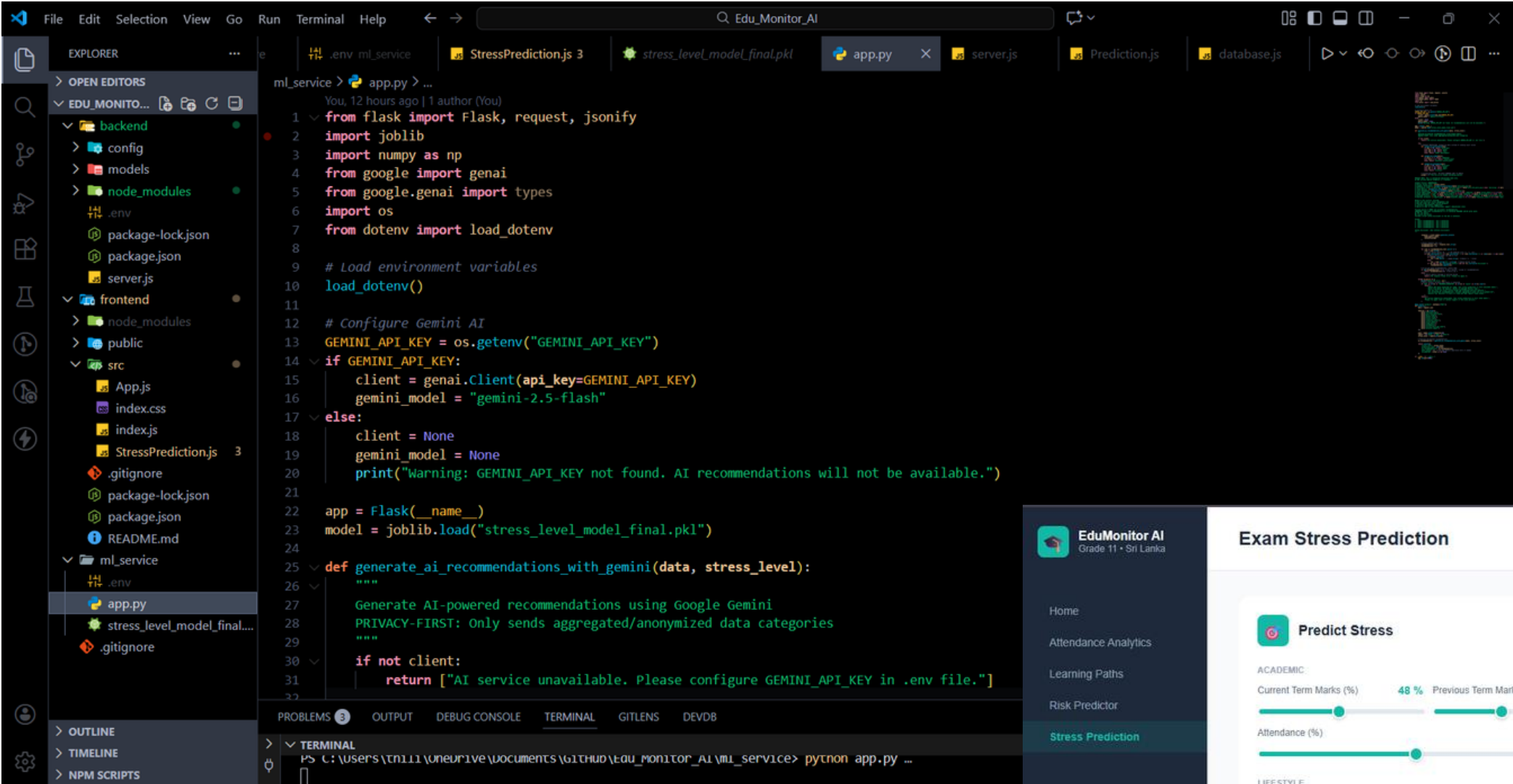
Design Excellence / Contribution

- Utilizes a high-accuracy Random Forest machine learning model for stress classification.
- Developed a modular architecture integrating ML, AI, and web technologies (MERN + Flask).
- Delivered a scalable stress monitoring component that enhances the overall Edu Monitor AI system.
- Enables early detection of students at risk of burnout or academic failure.

User Feedback on Prototype

- Users found the system easy to use and appreciated the simple data entry process.
- Users felt the stress explanations helped them understand their problems better.
- Students reported that the personalized recommendations were helpful and relevant to their real situations.
- Participants expressed confidence in the system because their personal data was protected.

PROJECT EVIDENCE



COMMERCIALIZATION

Commercialization / Sustainability

- Cloud-based SaaS platform for schools
- Scalable, multilingual, curriculum-flexible system
- Supports digital transformation & long-term educational value
- Continuous AI model improvements ensure sustainability

Customer Persona

- Schools
- Teachers
- Students

Market Size

- Government & Private schools
- High demand for AI-driven smart education solutions

Unique Value

- Subscription-based packages
- Institutional licensing
- Customization & integration fees
- Maintenance & support revenue

Cost Recovery

- All-in-one intelligent education ecosystem
- Supports differently-abled learners
- Real-time insights & decision support for teachers

The background features abstract geometric elements. In the top-left corner, there are overlapping light blue and dark blue organic shapes. In the top-right, a large blue organic shape contains two overlapping circles with diagonal stripes (one blue/white, one dark blue/white). In the bottom-left, a blue organic shape contains two overlapping circles with diagonal stripes (one blue/white, one dark blue/white). In the bottom-right, there are overlapping light blue and dark blue organic shapes. A thin dark blue line runs horizontally across the top and bottom of the page. Two vertical dark blue lines are positioned on the left and right sides, each intersected by a horizontal dotted line. The text "THANK YOU" is centered in a bold, dark blue, sans-serif font.

THANK YOU